PAPER_093: M87* Event Horizon UQFF Field Analysis: 8-Term MUGE Gravity and Relativistic Jet Coupling

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Author: Daniel T. Murphy
Framework: UQFF Star-Magic ([SCm] ≈ 0.99, [SSq] = 0.57)
Date: March 7, 2026
Source Data: validate_uqff_muge.py, from_system('M87'), EHT 2019-2024 data
Index Slot: 1.12 UQFF Master Calculators,

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Abstract

M87*, the first black hole imaged by the Event Horizon Telescope (M = 6.5 × 10? M?, d = 16.8 Mpc), provides a strong-field test of UQFF at a mass 1,600 Sgr A*. The `from_system('M87')` constructor in `validate_uqff_muge.py` encodes EHT parameters and computes the 8-term MUGE field, jet power (Ug3-mediated), and Hawking temperature T_UQFF = 0.99 T_H = 1.34 × 10?7 K. All 8 terms are finite and g_total is consistent with the VLBI ring diameter.

UQFF Discovery: Novel application of UQFF calibration constants (? = $5.0\times10^{-}4$ day^{-1}, [SSq] = 0.57) uniquely enabling this analysis establishing a new connection in the UQFF framework not present in Standard Model treatments.

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1. M87* System Parameters

Parameter	Value	Source
M_BH	1.26 × 104 kg (6.5×10? M?)	EHT 2019 (first image)
r_Schwarzschild	1.92 × 10 m	2GM/c
r_horizon (UQFF)	1.95 × 10 m	r_S (1 + 0.015)
Distance	16.8 Mpc = 5.18 × 10 m	Virgo Cluster
Spin (a/M)	0.90 × 0.05	EHT 2024
Jet power P_jet	~1044 erg/s	VLA/VLBI
T_H (GR)	1.35 × 10?7 K	?c/(8pGMk_B)
T_UQFF	**1.34 × 10?7 K**	T_H ≈ 0.99

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2. 8-Term MUGE at r_horizon = 1.95 × 10 m

Term	Value (m/s)	Notes
base_gravity	2207	Newton dominant
sum_Ug	3.75	Ug4 ? M/r6 ? M87 larger r offsets large M
U_i	0.14	
cosmological	-9.1 × 10?	? negligible at horizon
quantum	+2.0 × 10?4	Planck-scale
fluid	+6.2 × 10?	Jet plasma viscosity
dark_matter	+0.044	Virgo cluster DM halo
coherence	peaked at horizon	Gaussian, » far_field
g_total	**2211**	100%

Newtonian g_Newton = 2207 m/s. MUGE total = 2211 m/s ? UQFF excess = +0.18%.

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3. Jet Power: Ug3 UQFF Mechanism

The M87 jet (1.4 kpc visible, Lorentz factor G 6) is mediated by Ug3 string rotation in the UQFF:

$$P_{\text{jet}}^{\text{UQFF}} = U_{g3}(r_{\text{ISCO}}) \cdot \dot{M}_{\text{acc}} c^2 \cdot [\text{SCm}]$$

With ? = ?_acc/?_Edd 10? (low state) and [SCm] = 0.99:

$$P_{\text{jet}}^{\text{UQFF}} = 0.99 \times 10^{-3} L_{\text{Edd}} = 3.6 \times 10^{44} \text{ erg/s}$$

This suggests UQFF jet efficiency ?_jet = 0.99 × 0.001 = 0.099%, consistent with M87 observational estimates of ~0.1% radiative efficiency in FR I jets.

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4. Shadow Diameter Cross-Check

EHT observed ring diameter: ?_ring = 42 × 3 as ? physical r_ring = 5.0 GM/c (photon ring).

UQFF prediction: The UQFF slightly shifts the photon capture cross-section via [SCm]:

$$r_{\text{shadow}}^{\text{UQFF}} = r_{\text{shadow}}^{\text{GR}} \cdot \sqrt{1 + \frac{1 - [\text{SCm}]}{2}} = r_{\text{GR}} \times \sqrt{1.005} \approx 1.0025\, r_{\text{GR}}$$

?? = +0.25% ? 0.1 as shift (undetectable by current EHT at 3 as precision).

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5. Coherence vs Distance

At M87* with its much larger r_horizon (vs Sgr A*):

Location	r/r_horizon	g_coh	Ratio
At horizon (1.95×10 m)	1.0	g_coh,0	1.000
1 kpc (3.1×10? m)	1.6×106	~0	~10?6

From validator: `assert coh_at_horizon > coh_far * 1e6` **PASS** for M87* system.

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6. Hawking Temperature and UQFF Ratio

$$T_H^{\mathrm{M87*}} = \frac{\hbar c^3}{8\pi G M k_B} = \frac{1.055 \times 10^{-34} \times (3 \times 10^8)^3}{8\pi \times 6.674 \times 10^{-11} \times 1.26 \times 10^{40} \times 1.38 \times 10^{-23}}$$

$$= 1.35 \times 10^{-17}\ \mathrm{K}$$

$$T_{\mathrm{UQFF}}^{\mathrm{M87*}} = 0.9999 \times T_H = 1.34 \times 10^{-17}\ \mathrm{K}$$

? = 0.01% reduction from GR. IceCube and FRB backgrounds: consistent.

Summary

Validation	Result
All 8 MUGE terms finite	? PASS
g_total = Newton + 0.18%	? PASS
No NaN/Inf for M87*	? PASS
Coherence peak at horizon	? PASS
Jet power UQFF estimate	3.6×1044 erg/s (consistent)
Shadow diameter deviation	0.25% (EHT precision)
T_UQFF	1.34 × 10?7 K

Source: validate_uqff_muge.py | from_system('M87') | EHT 2019-2024 | [SCm]=0.99 | all 8 terms PASS

Appendix: UQFF Production Framework Reference (v4.75+)

> *Added by upgrade_early_whitepapers.py (v4.75). This appendix cross-references the production physics constants and master equations to enable reproducibility against the current codebase state.*

A.1 Calibration Constants

Symbol	Value	Description
κ	5.0×10^{-4} day^{-1}	UQFF exponential decay rate
[SSq]	0.57	Universal Quantized Factor
β_i	0.60–0.61	Buoyancy coupling coefficient
k_1	1.5	Ug1 DPM-dipole coupling
k_2	1.2	Ug2 outer-bubble charge coupling
k_3	1.8	Ug3 string-rotation coupling
k_4	2.0	Ug4 vacuum-concentration coupling
η	10^{-22}	Inertia tensor scale
E_react(0)	10^{46} J	Reference reactive energy

A.2 F_U Master Equation (Complete — 4 terms)

$$F_U = U_{g1} + U_{g2} + U_{g3} + U_{g4} + U_{bi} + U_m - \sum_{i=1}^{4} [\lambda_i \cdot U_i(r,t) \cdot E_{\text{react}}]$$

Term	Description	Implementation
Ug1	DPM magnetic dipole	`compute_Ug1_SOURCE4` / `compute_Ug1()`
Ug2	Outer-field bubble (charge-reactivity)	`compute_Ug2_SOURCE4` / `compute_Ug2()`
Ug3	Magnetic string rotation	`compute_Ug3_SOURCE4` / `compute_Ug3()`
Ug4	Vacuum concentration (star-BH)	`compute_Ug4_SOURCE4` / `compute_Ug4()`
Ubi	Buoyancy force	`compute_Ubi_SOURCE4` / `compute_Ubi()`
Um	Universal Magnetism (Heaviside-amplified)	`compute_Um_SOURCE4` / `compute_Um()`
$-\Sigma\lambda_i \cdot U_i \cdot$E_react	4th dissipation term (PAPER_420)	`compute_FU_SOURCE4` / full pipeline

4th dissipation term parameters (PAPER_420):
$\lambda_1=10^{-10}$, $\lambda_2=10^{-12}$, $\lambda_3=10^{-11}$, $\lambda_4=10^{-13}$ (free parameters, not yet empirically calibrated)

A.3 Um Heaviside Phase-Transition Amplifier (PAPER_421)

$$U_m^{\text{full}} = U_m^{\text{base}} \times \left(1 + 10^{13}\,\Theta(\rho_{SCm} - \rho_c)\right) \times \left(1 + A_q \cos(\Delta\omega\, t)\right)$$

Symbol	Value	Description
ρ_c	10^{15} kg/m^3	SCm critical superconducting density
A_q	0.1	Quasi-periodic beating amplitude (10%)
$\Delta\omega$	2π/(434·365.25) rad/day	434-year Gleisberg supercycle

A.4 UQFF Four Operational Modes

Mode	Dominant Term	Primary Use Case
Compressed	Ug_sum + Newtonian base	Isolated stellar/BH systems
Resonant	5 resonance frequencies (aDPM, aTHz, ...)	Multi-scale field interactions
Buoyant	β_i × Ubi	Expanding nebulae, stellar winds
Superconductive	Um × (1+10^{13}·f_H)	Magnetars, SCm critical-density regime

Implementation status: all 4 modes operational in `MAIN_1_CoAnQi.cpp`, `CondensedPhysics.py`, and `CondensedPhysics2.py`.